

Water Jug Problem (AI)

Aim

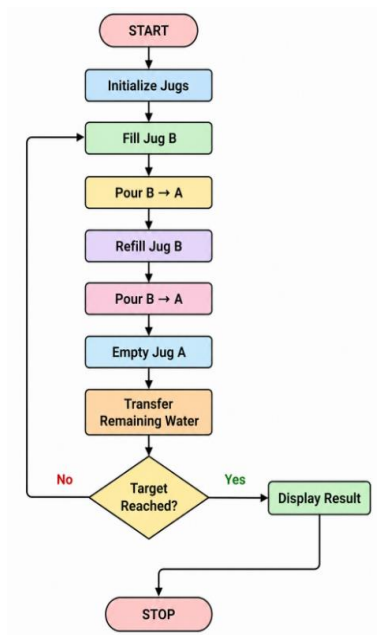
To solve the Water Jug Problem using Artificial Intelligence techniques and obtain the required quantity of water.

Objectives

- To understand state-space representation.
- To apply problem-solving techniques in AI.
- To achieve the target amount of water.
- To understand the concept of search algorithms.

Algorithm

1. Start.
2. Fill Jug B (3 liters).
3. Pour water from Jug B into Jug A.
4. Refill Jug B.
5. Pour water into Jug A until Jug A is full.
6. Empty Jug A.
7. Transfer the remaining water from Jug B to Jug A.
8. Check if Jug A contains 2 liters.
9. If yes, display the result.
10. Stop.



SOURCE CODE

```

b = jug2
print((a, b))

# Fill Jug A from Jug B
transfer = min(jug1 - a, b)
a += transfer
b -= transfer
print((a, b))

# Empty Jug A
a = 0
print((a, b))

# Transfer remaining water
a = b
b = 0
print((a, b))

# Fill Jug B again
b = jug2
print((a, b))

# Pour into Jug A
transfer = min(jug1 - a, b)
a += transfer
b -= transfer
print((a, b))

print("\nTarget Achieved: 2 liters in Jug A")

```

OUTPUT

```
= RESTART: C:/Users/ATHIN/AppData/Local/Temp/ipykernel_12345/12345.py
(Jug A, Jug B)
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
(0, 2)
(2, 0)
(2, 3)
(4, 1)

Target Achieved: 2 liters in Jug A
```

Result

The Water Jug Problem was successfully solved, and the required quantity of water (2 liters) was obtained using the sequence of valid operations.